

4.10 Antiderivatives

Motivation: Suppose a particle moving in straight line has velocity function $v(t) = t^2 + 2t$. Suppose $s(t)$ is the position function describing the motion of the particle. You are told that $s(0) = 0$. How might we find a formula for $s(t)$?

Definition: A function F is called the **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I .

Theorem: If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

Example 1: Find the most general antiderivative of each of the following functions.

a) $f(x) = \sin(x)$

b) $f(x) = x^n$ (where n is a real number)

c) $f(x) = \frac{1}{x}$

d) $f(x) = e^x$

Example 2: Find the most general antiderivative of each of the following functions.

a) $f(x) = \sec^2(x)$

b) $f(x) = 2^x \ln(2)$

c) $f(x) = \cos(x)$

d) $f(x) = \frac{1}{1+x^2}$

e) $f(x) = \csc(x) \cot(x)$

Theorem: If F and G are antiderivatives of f and g respectively, then

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Note: This theorem just highlights that we can use the Constant Multiple Rule and the Sum & Differences Rules in a “backwards” direction.

Example 3: Find the most general antiderivative of $f(x) = 6x^2 - 2x + 1$.

Example 4: Find the most general antiderivative of $g'(x) = 4\cos(x) + \frac{2x^5 - \sqrt{x}}{x}$.

Example 5: Find *the* function f if $f'(x) = 3x^5 - 2x^3 + 6x + 1$ and $f(1) = 3$.

Note: Problems like Ex#5 (and the motivating example) are called *initial value problems* since we are told a value of the original function.

Definition: An equation which includes a function and its derivative(s) is called a **differential equation**.

Ex: $y'' + 3y' + y = x$ & $\frac{dy}{dx} = 4y$ & $\frac{dy}{dx} = 6x + 2$

Engineering majors and some other majors require a course in differential equations. They will need to analyze a differential equation by finding a formula, a graph, or numerical estimates for a function y which satisfies the differential equation.

Example 6: Find f if $f'(x) = e^x + 20(1+x^2)^{-1}$ and $f(0) = -2$.

Example 7: A particle moves in a straight line and has acceleration given by $a(t) = 3t^3 + 3\cos(t)$. The velocity of the particle at time 0 is 7 *m/s* and its position at time 0 is 10 *meters* to the right of the origin. Find the position function $s(t)$ for the particle.